

STATIC WEB HOSTING

Here is the Static Website Hosting on S3 explained in simple step-by-step procedure (beginner to DevOps level).

What is Static Website Hosting in S3?

You can host a website that contains only:

HTML

CSS

JavaScript

Images

No backend server required.

Example: Portfolio, landing page, documentation site.

Step-by-Step Procedure (Simple)

Step 1: Create a Bucket

Login to AWS Console

Go to S3

Click Create bucket

Fill details:

Bucket name: must be unique

Example:

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siddhesh-website

Region: Choose nearest (Mumbai – ap-south-1)

Click Create bucket

Step 2: Disable Block Public Access.

By default, S3 blocks public access.

Open your bucket

Go to Permissions

Click Block public access

Click Edit

Uncheck:

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Block all public access

Save changes

Confirm by typing: confirm

Step 3: Upload Website Files

Prepare files:

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index.html

style.css

image.png

Upload:

Go to Objects

Click Upload

Add files

Click Upload

Step 4: Make Files Public

Select all uploaded files

Click Actions → Make public using ACL

OR (Recommended) use Bucket Policy.

Step 5: Add Bucket Policy (Recommended)

Go to: Permissions → Bucket Policy

Paste:

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```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicRead",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::siddhesh-website/*"
    }
  ]
}
```

Replace bucket name if needed.

Save.

This allows public to read files.

Step 6: Enable Static Website Hosting

Go to Properties

Scroll to Static website hosting

Click Edit

Select:

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Enable

Enter:

Index document: index.html

Error document: error.html (optional)

Save.

Step 7: Access Your Website

After enabling, you will get a URL:

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<http://siddhesh-website.s3-website-ap-south-1.amazonaws.com>

Open in browser → Your website is live 🎉

How It Works (Architecture)

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User Browser

|

v

S3 Website Endpoint

|

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Bucket → index.html, CSS, images

No EC2, no server.

Best Practices (DevOps Level)

1. Use CloudFront (Recommended)

Benefits:

HTTPS support

Faster global access

Security

Architecture:

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User → CloudFront → S3

2. Keep Bucket Private (Advanced)

Disable public access

Allow access only via CloudFront

3. Enable Versioning

So you can rollback website if needed.

4. Use Lifecycle

Move old files to cheaper storage.

Common Errors & Fix

Access Denied

Check:

Block public access disabled

Bucket policy correct

index.html not loading

Check:

Index document name exactly index.html